

LINKED LIST

PROBLEM 7

QUESTION:

DS	Session	Linked List	Question Information	Level 1	Challenge 37
Question description the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly. Dr.Ramprasath is a faculty, who handling data structure course for EEE department second year students. one day this faculty was handling very interesting topic in data structure such that Linked List, During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept. So the faculty decided to conduct test on the topic of Linked List. the question was given to last bench students that is, The nodes are deleted before a certain given node in the linked list. For example if the given Linked List is 5->10->15->20->25 and delete before 15 then the Linked List becomes 15->20->25. Constraint : $1 < N < 1000$ $1 < P < N-1$ INPUT Format First line contains the number of datas- N. Second line contains N integers(the given linked list).					

Third line contains position of the node to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
4 9 77 12 6 12	4 91 77 12 61 2
EXPECTED OUTPUT	EXPECTED OUTPUT
Linked List:->12->6	Invalid Node! Linked List:->91->77->12->61

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
struct node	node *next;	void create()

Test Case 4 KEYWORD for(i=0;i<n;i++)	Test Case 5 KEYWORD p1=p1->next	Test Case 6 KEYWORD void del()
Complexity Test Cases		
Test Case 1 CYCLOMATIC COMPLEXITY 3	Test Case 2 TOKEN COUNT 270	Test Case 3 NLOC 58

SOURCE CODE:

```

#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node *head = NULL;
int n;

void create() {
    struct node *tail = NULL;
    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        struct node *new_node = (struct node*)malloc(sizeof(struct node));
        new_node->data = val;
        new_node->next = NULL;
        if (head == NULL) {
            head = new_node;
        } else {
            tail->next = new_node;
        }
        tail = new_node;
    }
}

void del(int P) {
    struct node *curr = head;
    int found = 0;
    while (curr != NULL) {
        if (curr->data == P) {
            found = 1;
            break;
        }
        curr = curr->next;
    }

    if (!found) {
        printf("Invalid Node! ");
    } else {
        struct node *p1 = head;
        while (p1 != curr) {
            struct node *temp = p1;
            p1 = p1->next;
            free(temp);
        }
        head = curr;
    }
}

int main() {
    if (scanf("%d", &n) != 1) return 0;
    create();
    int P;
    scanf("%d", &P);
    del(P);
    printf("Linked List:");
    struct node *temp = head;
    while (temp != NULL) {
        printf("->%d", temp->data);
        temp = temp->next;
    }
    printf("\n");
    return 0;
}

```

OUTPUT:

```
4
9 77 12 6
12
Linked List:->12->6
```

```
4
91 77 12 61
2
Invalid Node! Linked List:->91->77->12->61
```